

Project 2 – Users and Groups

Contents

Background	1
Objectives	1
Prerequisites	1
User Management	2
Basic Group Management	3
More on users and groups	5
Submission	5

Background

We will work with users and groups in our Fedora server VM.

Objectives

1. Start Fedora server virtual machine.
2. Create new users. `[ useradd <user name>]`
3. Manage passwords. `[ passwd <user name>],`
`<enter password>.`
4. Create new groups. `[groupadd -g < group ID > < group name >]`
5. Modify groups. `[cat /etc/group]`
6. Delete groups. `[ groupdel <group name>]`
7. Create a new user and assign to a group. `[useradd <user name>],` __
and __`[ useradd -g <group name> -u <user ID> <user name> ]`__
8. Assign existing users to groups.
9. Document steps for inclusion in a project report to be submitted for review and grading in Blackboard.

Prerequisites

1. A Windows or MacOS device with a functioning hypervisor (e.g., VMware Workstation Pro).

User Management

Adding a user

1. Login as **root**, or use [su -]
2. Enter [**cat /etc/passwd**]. This command displays the contents of the password file where all users are defined.
3. Look at the last line in the passwd file, record the first column of it, which is the username van
4. Enter [**useradd**]. Watch the usage message displayed – you must at least specify a username!
5. Enter [**useradd Van**], where your name is the username, you want to add, such as your first name, or first character of your first name followed by your last name – if not already created by you before.
6. Display the [**cat /etc/passwd**] file again. Did you find your new username?
Yes
7. Enter [**passwd Van**], enter a password (e.g., **!Pa55w0rd**) of at least 6 characters containing at least one number and one special character.
8. *Re-enter the password* when prompted.
9. **Log out**, and then in as yourname.

More on creating user.

1. Login as **root** (regular user can't add a new user) , or use [su -]
2. Using the useradd command, add the user **jsmith**.
3. What command did you use: useradd jsmith
4. Assign **jsmith** the password of **!Pa55w0rd**.
5. What command did you use: [passwd jsmith]

6. **Log out** and **login** using **jsmith** to *verify* the username is created.

Changing user passwords

1. **Login** as **root**, or use [su -].
2. Using command *passwd* to change the password of user *jsmith* to **!Pa55w0rd**.
3. What command did you use: ____[**passwd jsmith**], _____
4. **Log out** and **login** using **jsmith** with the new password to confirm that it was changed.
5. Similarly, add another user **cbrown** with same password.
6. Enter [**userdel cbrown**] to delete the accounts for cbrown.
7. Check the /etc/passwd file (e.g., using [**cat /etc/passwd**]) to verify that the user was successfully removed so no such username exists in the file.
8. Try to **login** as **cbrown** – should fail!

Basic Group Management

Creating Groups

1. Login as **root**, or use [su -].
2. Enter [**cat /etc/group**] to view the file that defines the groups on the system.
3. Enter [**groupadd -g 2000 project1**] to create a group named *project1*. With *group id* 2000.
4. Now create *three* other groups on your system based on the following information:

Group name: cet349	GID: 2349
Group name: cet449	GID: 2449
Group name: cet549	GID: 2549
5. View the [**cat /etc/group**] file and note the new group additions. Notice that *there* are currently no users assigned to any of the groups.
6. **Log out**.

Modifying Groups

The `groupmod` command is the default method for modifying a group. The syntax is `groupmod <groupname>`.

1. Login as **root**, or use [su -].
2. Enter [`groupmod -n cet54900 cet549`] to *change cet549 to cet54900*. Enter [`cat /etc/group`] to *verify* the change. -n here means “new”.
3. Enter [`groupmod -g 2001 cet54900`] to change the group ID number to 2001. Enter [`cat /etc/group`] to *verify* the change. -g here means we are working on group id.
4. Now change the cet449 group to cet44900 with GID of 2002.
5. What command did you use : ____ [`groupmod -n cet44900 cet449`] ____
and ____ [`groupmod -g 2002 cet44900`] ____
6. Enter [`cat /etc/group`] to *verify* the change.
7. Now change the cet54900 group back to cet549.
8. What command did you use: ____ [`groupmod -n cet549 cet54900`] ____
9. Enter [`cat /etc/group`] to verify the change.

Deleting Groups

1. *Delete* group `cet349`.
2. What command did you use: ____ [`groupdel cet349`] ____
3. Check the [`cat /etc/group`] file to confirm the deletion.

Note, you can't delete the primary group of any user. To delete the group, you need to first change the user's primary group or delete the user. You should also check files to make sure no files are left with the deleted group's GID. For cet349, cet449 and cet549 we have not added any user to them.

More on users and groups

Create a new user and assign it to a group.

1. Login as **root**, or use [su -].
2. Use **useradd** command to create a user **ga** with user ID 512 and assign it to group **cet549**.
3. What command did you use: ____[**useradd ga**], ____
and ____[**useradd -g cet549 -u 512 ga**]_____
4. Add password **!Pa55w0rd** to user **ga**.
5. What command did you use: _____[**passwd ga**]_____

Assigning existing users to groups

1. Login as **root**, or use [su -].
2. Now add a user **test** with password **!Pa55w0rd**.
3. Now add user **test** to a few groups using the command: **usermod** by entering

```
usermod -G cet549,cet44900 test
```

*Note: -G add supplemental groups to a user. -g, on the other hand, work on primary group. Also, there is **no space** between **cet549** and **cet44900**.*

4. Now check the user information of **test** by entering

```
id test
```

What are the groups that user **test** is currently in?

Submission

Use this document (or one of your choosing) to submit your screen shots and any other relevant data. At a minimum, you should show the commands run after logging in as root. Please ensure your submission includes your name, class, date, and reference to this lab.

